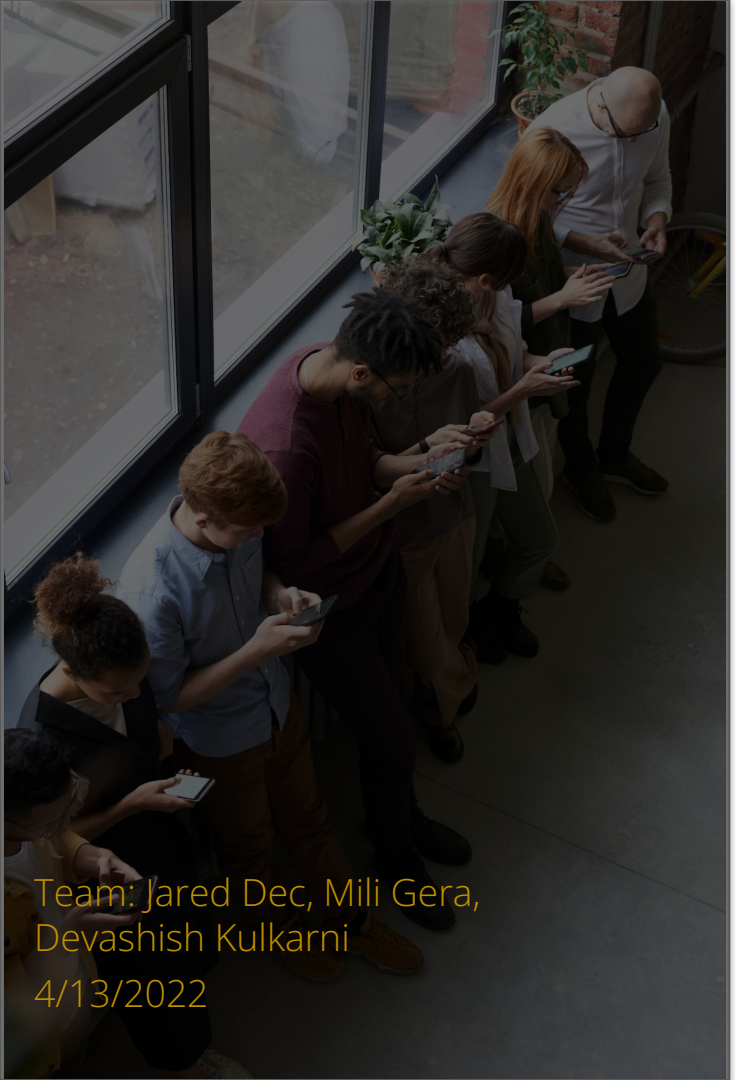

Hate Speech Detection
Hinglish and Korean Social Media

기자: 저는 이국주가 싫어요

ममता मौत की सौदागर है।

킹기리는요? 연애따로 결혼따로

@miss_dynamic__ तारीफ तो करो😐😐

पूरी दुनिया हैरान है ये जानकर भारत सब्सिडी पर सूअर पालता है

조용필 나대지마라

연기 실감나게 잘하시더군요

@sunilyadv फेक् है इस लिए

Team: Jared Dec, Mili Gera, Devashish Kulkarni

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Problem Statement

"Online hate and extremist narratives have been linked to abhorrent real-world events, including a current surge in hate crimes and an alarming increase in youth suicides that result from social media vitriol; … Social media platforms seem to be losing the battle against online hate and urgently need new insights" [1]

Current State

Research Conducted

463 papers found in systematic review of all hate speech detection research[2]

51% of the research conducted in English [2]

Research Needed

Many languages in the world some of which diverge both linguistically and culturally from English

Data/Chosen Languages

Korean

No linguistic relation to English

Culturally distant in hate-expression

Subtle: “설마 ㅈ[sic] 현정 작가 아니지??”. This Translates to , “There’s no way Hyeonjeong is really a writer, is there?”.

News entertainment articles (7,292 examples with 80:10:10 train, development, test split)

Hindi/Hinglish

Linguistic relation to English

Culturally similar in hate-expression

Direct (often): “हे प्रभु!! निंदा????? नेताओं की , और सरकारों के निकम्मेपन को छुपाने के लिए एक शब्द काफी है "निंदा" I”. This translates to, “Oh God!! Condemnation????? One word is enough to hide the inefficiency of politicians and governments”.

Twitter data (5,294 examples with 80:10:10 train, development, test split)

Research Objective

Compared the applicability of various transformer based language models on two linguistically different languages (as opposed to maximizing a single metric like an F1 score). Our languages of interest were Korean and Hindi

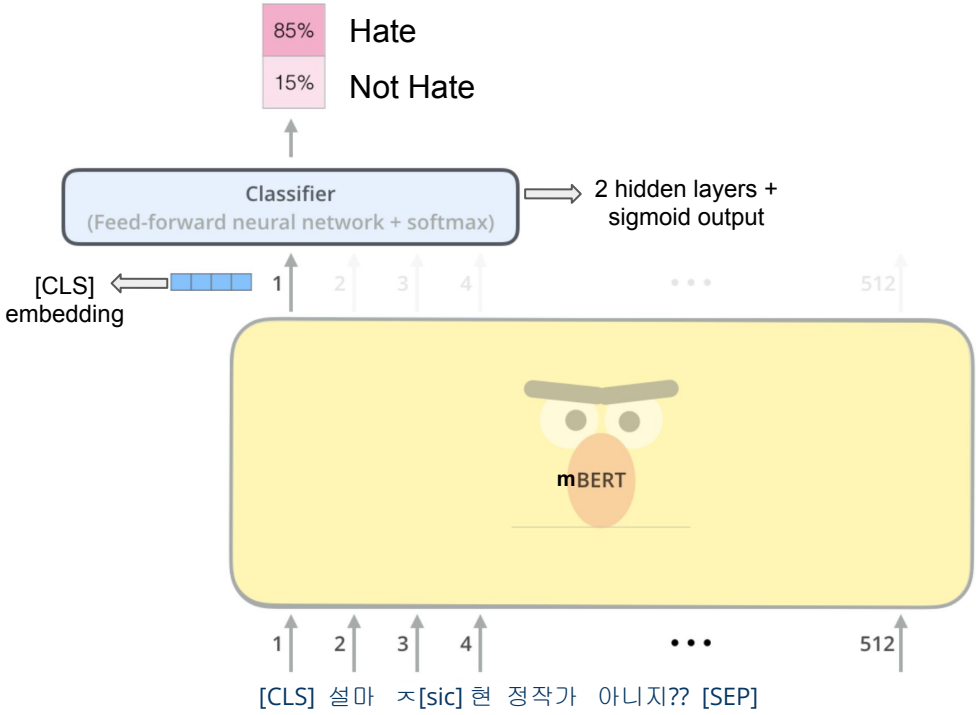
Initial Hypothesis

Due to cultural and linguistic variations (especially the subtleties seen in Korean hate speech), different model architectures may prove to be the “best-fit” models for Hindi and Korean. Specifically we hypothesized that Korean may see a bigger boost from the chosen mBERT baseline if a language-agnostic framework was used.

Models Used

mBERT[3] - baseline

- Identical architecture and training task as BERT, uses top 100 languages with the largest Wikipedia's
- Korean - 23rd, 585k articles (English - 6M articles)
- Hindi - 57th, 150k articles
- Used as a baseline for most work in the literature



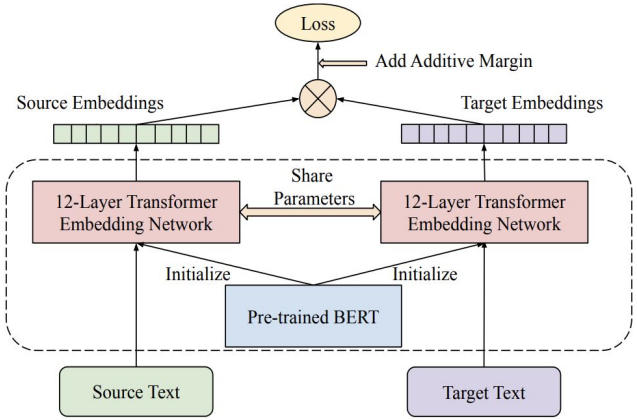
Hyperparameters

- **Best Performing Settings**
  - High learning rate
  - Largest hidden layers
  - Freeze mBERT layers
  - 0 dropout (Hindi), .15 dropout (Korean)
- **Pre-processing tweets-Reduced performance**
  - Replace username with special token
  - Remove URLs
  - Segment English Hashtags

mBERT Image Source:
"The Illustrated BERT, ELMo, and co. (How NLP Cracked Transfer Learning)", Jay Alammar, https://jalammar.github.io/illustrated-bert/

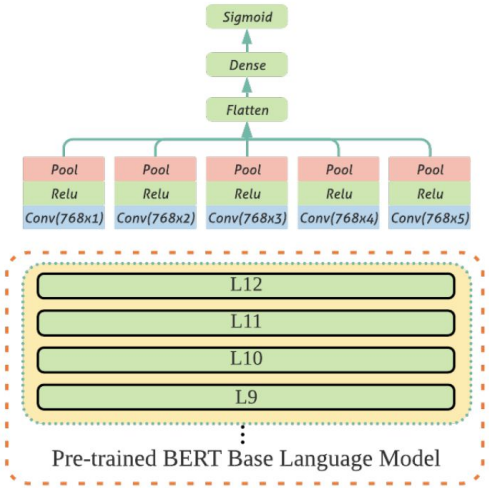
Models Used

LaBSE[4]



- mBERT-type model: does not perform well in mapping multilingual sentences of the same meaning to the same vector space[6]
- Advantageous to have cross-lingual sentence embeddings which "map" similar meaning sentences from two different languages?

LaBSE with CNN[5]



- mBERT-type model: create contextualized embeddings
- Layered with a CNN model: which extracts important fragments
- Does this give a bigger boost to the more subtle Korean hate speech data?

LaBSE Image Source:
"Language-Agnostic Bert Sentence Embedding", Yang, Yinfei, and Fangxiaoyu Feng, https://arxiv.org/pdf/2007.01852.pdf

LaBSE with CNN Image Source: KUISAIL at SemEval-2020 Task12: BERT-CNN for Offensive Speech Identification in Social Media, Safaya, A., Abdullatif, M. & Yuret D, https://arxiv.org/abs/2007.13184

Results

Recall Metric Prioritized When F1 Comparable

Strong baseline can be achieved using the mBERT model for both Korean and Hindi.

The marginal improvement with LaBSE also came at the expense of a higher false negative rate (lower recall)

Hindi FN Analysis

Hindi	TP	FP	TN	FN	Pr[1]	Re[2]	F1[3]
mBERT baseline	229	151	104	46	0.60	0.83	0.70
LabSE baseline hyper-parameters	189	63	192	86	0.75	0.69	0.72
LaBSE optimized hyper-parameters	192	65	190	83	0.75	0.70	0.72
LaBSE with CNN	189	70	185	86	0.73	0.69	0.71

1. Pr - Precision Rate, 2. Re - Recall Rate, 3. F1 - F1 Score (test)

30% FNs shared between mBERT and all LaBSE models

13% of the examples mBERT model predicted as FNs were predicted as TPs by all the LaBSE models

50% better overall performance by mBERT on the detection of FNs

Korean FN Analysis

Korean	TP	FP	TN	FN	Pr[1]	Re[2]	F1[3]
mBERT baseline	299	194	175	62	0.61	0.83	0.70
LaBSE baseline hyper-parameters	219	100	269	142	0.69	0.61	0.64
LaBSE optimized hyper-parameters	260	128	241	101	0.67	0.72	0.69
LaBSE with CNN	252	99	270	109	0.72	0.70	0.71

1. Pr - Precision Rate, 2. Re - Recall Rate, 3. F1 - F1 Score (test)

42% FN shared between mBERT and all LaBSE models

39% better overall performance by mBERT on detection of FNs

Highest F1

Highest Recall

Conclusions

- Our initial hypothesis could not be proven as both Hindi and Korean performed best on baseline mBERT
    - Language Agnostic model provides slight improvement over the baseline mBERT on F1 Score, but with a high cost on recall
- Large variations in prediction of individual false negative examples between LaBSE models

Hindi
- 55% disagreement on FN prediction among LaBSE models
- No obvious patterns as to why certain LaBSE models turn up TPs while others predict FNs

Korean
- 61% disagreement on FN prediction among LaBSE models
- Did notice that LaBSE missed many blatantly obvious hate speech examples. A possible reason could be use of highest level of formality. In that sense, LaBSE's could be capturing subtleties which increase FNs instead of decreasing them.

판사 음주운전 차량에 치어 뒤지길 빕니다" which roughly means "I hope the judge gets run over by a drunk driver" [but in very formal manner, which is likely sarcastic]

Other attempts

- **Zero-shot**
  - (train on English, test on Hi/Ko) -> Very poor performance
- **Transfer Learning**
  - (train on English, train again on Hi/Ko, test on Hi/Ko) -> Marginally Better than baseline for Hindi, worse for Korean
- **Translations**
  - (train on English, test on Hi/Ko translated to En) -> Very poor performance
  - (train on English, train again on Hi/Ko translated to En, test on Hi/Ko translated to En) -> Marginally Better than baseline for Hindi, worse for Korean

Future work

- Character-based models for noisy twitter data/typos (for example, ByT5)
- Investigate tokenizer to see patterns between Hate/Not hate text
- Comparison with other languages

References

[1] Johnson, N. "Hidden Resilience and Adaptive Dynamics of the Global Online Hate Ecology." *Nature*, 21 Aug. 2019, https://www.nature.com/articles/s41586-019-1494-7#ref-CR7

[2] Jahan, Md Saroar & Oussalah, Mourad. "A systematic review of Hate Speech automatic detection using Natural Language Processing." In ArXiv (2021), https://arxiv.org/pdf/2106.00742.pdf

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[4] Yang, Yinfei, and Fangxiaoyu Feng. "Language-Agnostic Bert Sentence Embedding." Google AI Blog, Google Research, 18 Aug, 2020, https://arxiv.org/pdf/2007.01852.pdf

[5] Safaya, A., Abdullatif, M. & Yuret D. "KUISAIL at SemEval-2020 Task12: BERT-CNN for Offensive Speech Identification in Social Media."arXiv:2007.13184 (2020), https://arxiv.org/abs/2007.13184

[6] Nurmanbetov, Daulet. "BERT Model Embeddings Aren't as Good as You Think - Towards Data Science." Medium, 14 Dec. 2021, towardsdatascience.com/cutting-edge-bert-nlp-model-bb0bfc8b7aec